APPLICATION OF OPTIMAL CONTROL THEORY TO DYNAMIC SOARING OF SEABIRDS

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Abstract: Optimal control theory is applied as a method for determining the minimum wind strength required for dynamic soaring of seabirds. Dynamic soaring is a flight technique by which seabirds extract energy from shear wind existing in an altitude layer close to the water surface. Mathematical models for describing the soaring motion of a bird and for the shear wind are presented. Optimality conditions are formulated using the minimum principle. Switching conditions are introduced to deal with a state constraint. Numerical results of high accuracy are generated using an efficient computational procedure based on the method of the multiple shooting for an albatross as a representative for seabirds performing dynamic soaring.

NOMENCLATURE

a_{ki}	abbreviation factor
C_D	drag coefficient
C_L	lift coefficient
D	drag
g	acceleration due to gravity
H	Hamiltonian

h	altitude
J	performance criterion
k	drag factor for describing lift effect
L	lift
m	mass of bird
S	reference area
t	time
u_{Kg}, v_{Kg}, w_{Kg}	speed components
V	airspeed
V_K	inertial speed
V_W	wind speed
x_K, y_K, z_K	geodetic coordinate system
χ_a	flight azimuth angle
γ_a	flight path wind angle
λ_i	Lagrange multiplier
μ_a	flight bank wind angle

1. INTRODUCTION

Dynamic soaring is a flight method by which an object in gliding flight (bird, sailplane) extracts energy from horizontally moving air. The possibility of extracting energy for continuous dynamic soaring requires that the horizontally moving air is non-uniform. This means that the horizontal wind speed changes with altitude. Such a type of wind is called shear wind or shear flow.

Dynamic soaring is observed with seabirds which utilize the shear flow in an altitude region close to the water surface. Here, the wind speed rapidly increases in a small altitude interval termed boundary layer, from zero to the value of the free air flow. There are several seabirds that perform dynamic soaring. Among these, the albatross is the most famous and considered the master of dynamic soaring (Ref. 1).

The possibility of utilizing shear wind for soaring flight and the basic mechanism of the energy transfer from the moving air to the bird have been early considered and clarified, Refs. 1-4. Since then, dynamic soaring is an issue of continuous interest and the knowledge has been continually increased, Ref. 5. Investigations on energy estimations and numerical simulations were performed, yielding an improved understanding of dynamic soaring. The research includes papers which are concerned with mathematical treatments of dynamic soaring, Refs. 6-11. Other papers are more dealing with ornithological aspects, Refs. 12-14. Modern optimization techniques have been applied to the dynamic soaring problem, yielding results on the minimum required wind strength, Refs. 15-17. Recent experimental research is concerned with tracking of albatrosses, providing results on their enormous flight performance, Refs. 18-23.

It is purpose of this paper to present a rigorous mathematical treatment concerned with dynamic soaring of seabirds. This relates to the mathematical model for describing the motion of the birds and to the optimization method to generate solutions for the minimum wind strength required for dynamic soaring. Numerical results are presented showing the form of the optimal dynamic soaring trajectory requiring minimum wind strength and the properties of state and control variables for achieving this goal.

2. BASIC CONSIDERATIONS ON DYNAMIC SOARING

Dynamic soaring comprises a rather complex and a highly dynamic flight maneuver involving a complicated control structure and a corresponding behavior of the state variables. Consequently, no simple and direct access to this problem is possible to yield closed-form solutions. By contrast, there are other soaring techniques which consist of rather simple flight maneuvers to enable an energy gain for the bird. These problems can be solved with direct approaches. Such soaring flights are thermalling and hang gliding in up-wind fields which are utilized by birds or sailplanes. An illustration is given in Fig. 1 which shows flight conditions in up-wind fields. Basically, the flying object is moved upwards by the air, resulting in a corresponding energy increase. This energy source can be utilized with rather simple maneuvers, like circling or even straight motions representing basically steady-state flight conditions. Correspondingly, comparatively simple solutions are known for such problems, Ref. 24.

By contrast, dynamic soaring consists of a complex and unsteady flight maneuver for transferring energy to the flying object from the moving air which is not moving upwards, but horizontally. A dynamic soaring flight maneuver is illustrated in Fig. 2. There is a horizontal wind which shows a shear flow characteristic. The wind rapidly increases from very small values immediately above the sea surface to the free air flow speed. For transferring energy from the moving air to the bird from such a shear wind, a complex

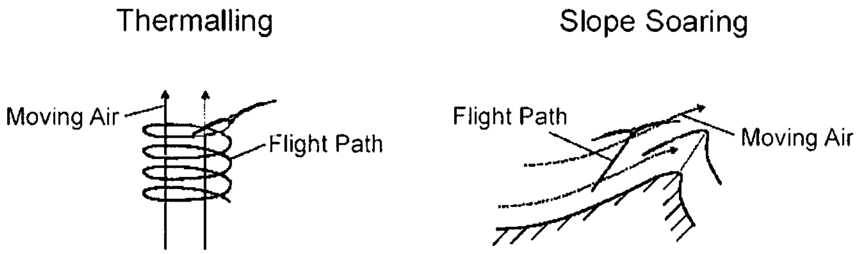

Figure 1. Flight paths and energy transfer in up-wind regions

flight maneuver as illustrated in Fig. 2 is necessary. It basically consists of four phases 1 to 4, yielding:

1) Phase 1
   Lower curve close to sea surface (change of flight direction from leeward to windward)
2) Phase 2
   Climb (windward flight)
3) Phase 3
   Upper curve (change of flight direction from windward to leeward)
4) Phase 4
   Descent (leeward flight)

Phases 1 to 4 form a cycle which is the basic constituent of dynamic soaring. By periodically repeating the dynamic soaring cycle, the bird can perform a range flight without flapping its wings.

With a dynamic soaring flight maneuver as shown in Fig. 2, it is possible for the bird to attain an energy gain from the moving air. This enables it to achieve an enormous flight performance. An example for the achievable performance is given in Fig. 3 which presents the trajectory of an albatross tracked by satellite measurements. The length of the trajectory shown in Fig. 3 is 6479 km which the bird has traveled in 8.15 days.

Focus of this paper is on the minimum shear wind strength required for dynamic soaring. There are dynamic soaring trajectories which show the same energy state of the bird at the end of a dynamic soaring cycle as at its beginning. These trajectories can be designated as energy-neutral. The designation "energy-neutral" means that the energy gain from the moving air is just sufficient to compensate for the energy loss due to drag after completing a dynamic soaring cycle. There is a great variety of energy-neutral trajectories. One of these is of particular concern: It is the one which requires the minimum shear wind strength. Having knowledge of this energy-neutral trajectory, it is possible to judge whether or not the shear

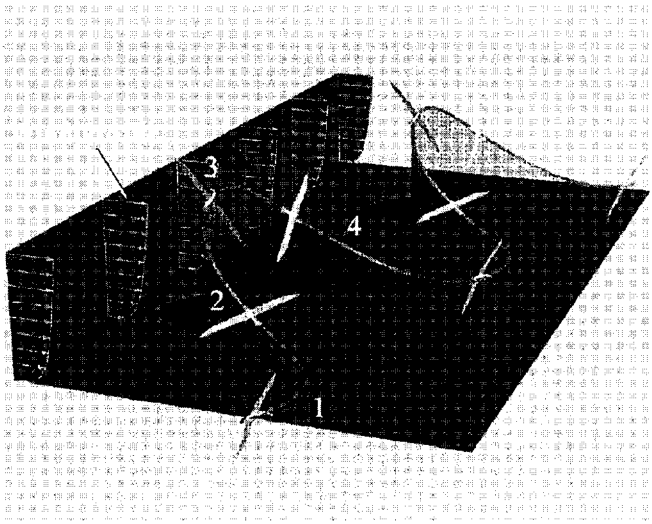

Figure 2. Dynamic soaring trajectory and energy transfer

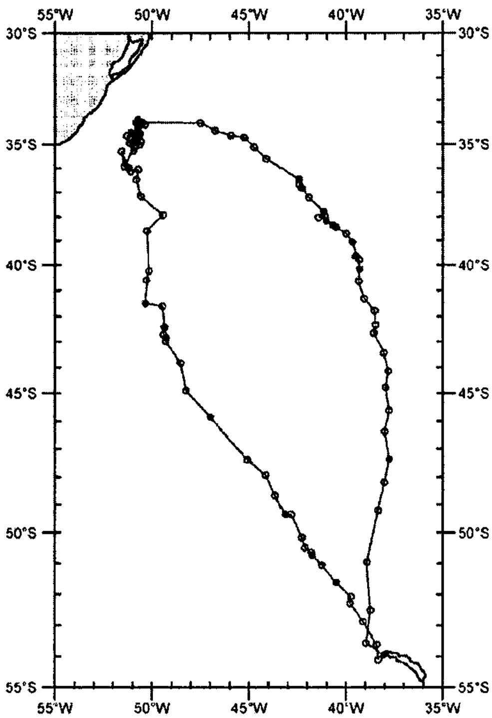

Figure 3. Flight path of an albatross (from Ref. 19)

wind strength in the areas of the seabirds in mind is sufficient for dynamic soaring.

3. MATHEMATICAL MODEL FOR DESCRIBING THE MOTION OF THE BIRD

The dynamics of birds in soaring flight can be described using a point mass model. Reference is made to an earth fixed coordinate system, and the moving air is appropriately accounted for. Fig. 4 shows the earth fixed reference system and the speed vectors describing the moving air and the motion of the bird. The equations of motion may be expressed as

$$
\begin{aligned}
\frac{\mathrm{d}u_{Kg}}{\mathrm{d}t} &= -a_{u1}\frac{D}{m} - a_{u2}\frac{L}{m} \\
\frac{\mathrm{d}v_{Kg}}{\mathrm{d}t} &= -a_{v1}\frac{D}{m} - a_{v2}\frac{L}{m} \\
\frac{\mathrm{d}w_{Kg}}{\mathrm{d}t} &= -a_{w1}\frac{D}{m} - a_{w2}\frac{L}{m} + g \\
\frac{\mathrm{d}x_{Kg}}{\mathrm{d}t} &= u_{Kg} \\
\frac{\mathrm{d}y_{Kg}}{\mathrm{d}t} &= v_{Kg} \\
\frac{\mathrm{d}h}{\mathrm{d}t} &= -w_{Kg}
\end{aligned}
\tag{1}
$$

where the coefficients a_{ij} denote functions of the path angles χ_a, γ_a and μ_a, given by the following relations

$$
\begin{aligned}
a_{u1} &= \cos\gamma_a \cos\chi_a \\
a_{u2} &= \cos\mu_a \sin\gamma_a \cos\chi_a + \sin\mu_a \sin\chi_a \\
a_{v1} &= \cos\gamma_a \sin\chi_a \\
a_{v2} &= \cos\mu_a \sin\gamma_a \sin\chi_a - \sin\mu_a \cos\chi_a \\
a_{w1} &= -\sin\gamma_a \\
a_{w2} &= \cos\mu_a \cos\gamma_a
\end{aligned}
\tag{2}
$$

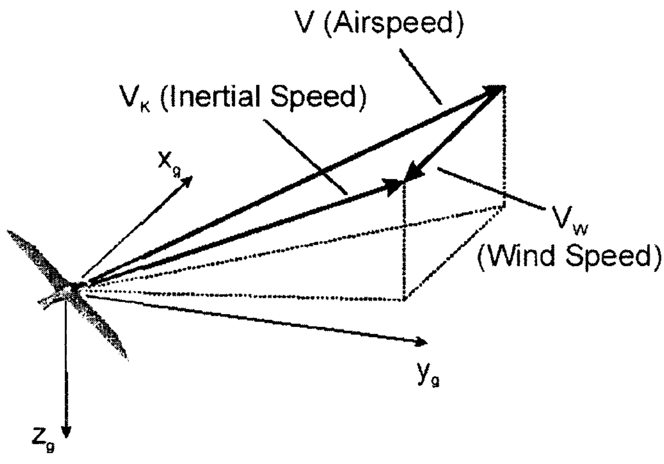

Figure 4. Coordinate system x_g, y_g, z_g and speed vectors V_K, V, V_w for describing the flight in a horizontal shear wind (V_K: inertial speed measured relative to the ground, V: airspeed of the bird relative to the moving air, V_w: wind speed). The x_g axis is aligned with the wind speed vector V_w, the z_g axis points vertically downward.

The aerodynamic forces are drag and lift which read

$$\begin{aligned} L &= C_L(\rho/2)V^2S \\ D &= C_D(\rho/2)V^2S \end{aligned} \tag{3}$$

The drag characteristics may be modeled as

$$C_D = C_{D0} + kC_L^2 \tag{4}$$

where the lift coefficient C_L is a control which is determined by optimality conditions described in a subsequent section.

The aerodynamic forces are dependent on airspeed $\vec{V}$, while the motion of the bird with regard to the earth is described by the inertial speed $\vec{V}_K = (u_{Kg}, v_{Kg}, w_{Kg})^T$. They are related to each other by the following expression

$$\vec{V} = \vec{V}_K - \vec{V}_W \tag{5}$$

With the use of $\vec{V}_W = (-V_W, 0, 0)^T$, Eq. (5) yields

$$\vec{V} = (u_{Kg} + V_{W,} v_{Kg,} w_{Kg})^{\mathrm{T}}$$
$$V = \sqrt{(u_{Kg} + V_W)^2 + v^2{}_{Kg} + u^2_{Kg}} \tag{6}$$

Two angles of the aerodynamic coordinate system used in Eqs. (1) and (2) are given by

$$\sin \gamma_a = -\frac{w_{Kg}}{V}$$
$$\tan \chi_a = \frac{v_{Kg}}{u_{Kg} + V_W} \tag{7}$$

The remaining angle μ_a which describes the banking of the lift vector is a control. It is determined by optimality conditions described in a subsequent section.

For a cycle of an energy-neutral trajectory, the following boundary conditions implying periodicity hold

$$u_{Kg}(0) = u_{Kg}(t_{cyc}), \quad v_{Kg}(0) = v_{Kg}(t_{cyc}), \quad w_{Kg}(0) = w_{Kg}(t_{cyc}), \quad h(0) = h(t_{cyc}) \tag{8}$$

where t_{cyc} describes the time at the end of a cycle. With an appropriate choice of the coordinate system, the boundary values of the longitudinal and lateral coordinates read

$$x_g(0) = 0, \quad x_g(t_{cyc}) : \text{free}$$
$$y_g(0) = 0, \quad y_g(t_{cyc}) : \text{free} \tag{9}$$

Control variables are the lift coefficient C_L and the bank angle μ_a. The lift coefficient is subject to the following constraint relation

$$C_{L\min} \leq C_L \leq C_{L\max} \tag{10}$$

4. MATHEMATICAL SHEAR WIND MODEL

The dynamic soaring of seabirds is possible because there is a shear wind at the sea. The shear wind is due to boundary layer effects of the moving air. This is illustrated in Fig. 5 which shows the shear wind profile (wind speed vs. altitude) for the altitude region of concern for the dynamic soaring of sea birds. From zero or very small values immediately above the sea surface, the wind speed rapidly increases and approaches the value of the free air flow.

There are various models for describing the shear wind characteristics. For the shear wind above the water surface, logarithmic or exponential models are used (Refs. 6,7,13,25). They may be expressed as

$$
\begin{aligned}
V_W &= V_{W\,ref}\left(\frac{h}{h_{ref}}\right)^p \\
V_W &= \bar{V}_W \frac{\ln(h/h_0)}{\ln(\bar{h}/h_0)}
\end{aligned}
\tag{11}
$$

The quantities $V_{W\,ref}$, h_{ref} and p as well as $\bar{V}_W$, h_0 and $\bar{h}$, denote reference values which are used for indicating the strength of the shear wind and for taking properties of the surface into account. The exponential model is applied in this paper, with $p = 0.143$.

5. OPTIMALITY CONSIDERATIONS

For the performance criterion, designated by J, the quantity $V_{W\,ref}$ can be used. This is because $V_{W\,ref}$ is a measure for the shear wind strength, yielding

$$
J = V_{W\,ref} \tag{12}
$$

The optimal control problem can then be formulated as to determine the controls, the initial conditions $\vec{V}_K(0) = (u_{Kg}(0), v_{Kg}(0), w_{Kg}(0))^T$ and $h(0)$ and the optimal cycle time t_{cyc} which minimize the performance criterion $J = V_{W\,ref}$ subject to the dynamic system Eq. (1), the boundary conditions Eqs. (8), (9) and the control constraints Eq. (10).

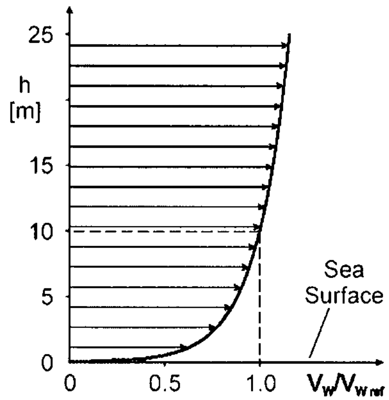

Figure 5. Model for describing shear wind in boundary layer above sea surface. The quantity V_W ref denotes the wind speed at the reference altitude for which a value of h = 10 m is chosen.

The optimal control problem described is solved with use of the minimum principle. For this purpose, the Hamiltonian is introduced

$$H = -\frac{a_{u1}D + a_{u2}L}{m}\lambda_u - \frac{a_{v1}D + a_{v2}L}{m}\lambda_v + \left(g - \frac{a_{w1}D + a_{w2}L}{m}\right)\lambda_w + u_{Kg}\lambda_x + v_{Kg}\lambda_y - w_{Kg}\lambda_h \tag{13}$$

with Lagrange multipliers

$$\lambda = (\lambda_u, \lambda_v, \lambda_w, \lambda_x, \lambda_y, \lambda_h)^{\mathrm{T}} \tag{14}$$

adjoined to the system Eq. (1). The following relations hold for the Lagrange multipliers

$$
\begin{aligned}
\frac{\mathrm{d}\lambda_u}{\mathrm{d}t} &= \frac{\partial}{\partial u_{Kg}}(a_{u1}D + a_{u2}L)\frac{\lambda_u}{m} + \frac{\partial}{\partial u_{Kg}}(a_{v1}D + a_{v2}L)\frac{\lambda_v}{m} + \\
&\qquad \frac{\partial}{\partial u_{Kg}}(a_{w1}D + a_{w2}L)\frac{\lambda_w}{m} - \lambda_x \\
\frac{\mathrm{d}\lambda_v}{\mathrm{d}t} &= \frac{\partial}{\partial v_{Kg}}(a_{u1}D + a_{u2}L)\frac{\lambda_u}{m} + \frac{\partial}{\partial v_{Kg}}(a_{v1}D + a_{v2}L)\frac{\lambda_v}{m} + \\
&\qquad \frac{\partial}{\partial v_{Kg}}(a_{w1}D + a_{w2}L)\frac{\lambda_w}{m} - \lambda_y \\
\frac{\mathrm{d}\lambda_w}{\mathrm{d}t} &= \frac{\partial}{\partial w_{Kg}}(a_{u1}D + a_{u2}L)\frac{\lambda_u}{m} + \frac{\partial}{\partial w_{Kg}}(a_{v1}D + a_{v2}L)\frac{\lambda_v}{m} + \\
&\qquad \frac{\partial}{\partial w_{Kg}}(a_{w1}D + a_{w2}L)\frac{\lambda_w}{m} - \lambda_h \\
\frac{\mathrm{d}\lambda_x}{\mathrm{d}t} &= 0 \\
\frac{\mathrm{d}\lambda_y}{\mathrm{d}t} &= 0 \\
\frac{\mathrm{d}\lambda_h}{\mathrm{d}t} &= \left[\frac{\partial}{\partial V_W}(a_{u1}D + a_{u2}L)\frac{\lambda_u}{m} + \frac{\partial}{\partial V_W}(a_{v1}D + a_{v2}L)\frac{\lambda_v}{m} + \right. \\
&\qquad \left. \frac{\partial}{\partial V_W}(a_{w1}D + a_{w2}L)\frac{\lambda_w}{m}\right]\frac{\mathrm{d}V_W}{\mathrm{d}h}
\end{aligned}
\tag{15}
$$

Because of periodicity properties, the following boundary conditions hold

$$
\lambda_u(0) = \lambda_u(t_{cyc}), \quad \lambda_v(0) = \lambda_v(t_{cyc}), \quad \lambda_w(0) = \lambda_w(t_{cyc}) \tag{16a}
$$

Further to the boundary conditions

$$
\lambda_x(t_{cyc}) = 0, \quad \lambda_y(t_{cyc}) = 0, \quad \lambda_h(t_{cyc}) = -1 \tag{16b}
$$

The Hamiltonian is constant because the system described by Eq. (1) is autonomous (i.e., not explicitly dependent on t). Furthermore, the time at the end of a cycle, t_{cyc}, is treated as free, yielding

$$
H = 0 \tag{17}
$$

The optimal controls are such that the Hamiltonian is minimized. From $\partial H / \partial \mu_a = 0$, it follows for the optimal bank angle control

$$\tan(\mu_a)_{opt} = \frac{\lambda_u \sin\chi_a - \lambda_v \cos\chi_a}{\lambda_u \sin\gamma_a \cos\chi_a + \lambda_v \sin\gamma_a \sin\chi_a + \lambda_w \cos\gamma_a} \tag{18}$$

Similarly from $\partial H / \partial C_L = 0$ for the optimal lift coefficient

$$(C_L)_{opt} = -\frac{1}{2k}\left[\frac{\lambda_u(\cos\mu_a \sin\gamma_a \sin\chi_a - \sin\mu_a \cos\chi_a)}{\lambda_u \cos\gamma_a \cos\chi_a + \lambda_v \cos\gamma_a \sin\chi_a - \lambda_w \sin\gamma_a} + \frac{\lambda_v(\cos\mu_a \sin\gamma_a \sin\chi_a - \sin\mu_a \cos\chi_a) + \lambda_w \cos\mu_a \cos\gamma_a}{\lambda_u \cos\gamma_a \cos\chi_a + \lambda_v \cos\gamma_a \sin\chi_a - \lambda_w \sin\gamma_a}\right] \tag{19}$$

Otherwise, the constraining bounds given by $C_{L\min}$ and $C_{L\max}$, Eq. (10), become active.

6. ALTITUDE CONSTRAINT AND SWITCHING CONDITIONS

The altitude range for the dynamic soaring maneuver has a lower limit given by the water surface. Therefore, it is necessary to introduce an altitude limit described as $h_{\min}$. This results in an altitude constraint given by

$$h \geq h_{\min} \tag{20}$$

As a consequence, there are additional optimization conditions which are presented in the following according to Ref. 26.

Basically, the altitude constraint can become active in two ways. First, a point of contact can exist at which the dynamic soaring trajectory touches the altitude limit $h_{\min}$. Second, there can be an arc on which the dynamic soaring trajectory stays for a finite interval. The second possibility occurred in the computational treatment of the optimization problem so that it will be considered in the following.

For treating the altitude constraint problem, the relation

$$G(h) = h_{\min} - h \leq 0 \tag{21}$$

is introduced. Calculating successively the time derivatives of G until an expression is obtained which is explicitly dependent on the control variables, the following result is obtained

$$
\begin{aligned}
&G^{(1)}(w_{Kg}) = w_{Kg} \\
&G^{(2)}(u_{Kg}, v_{Kg}, w_{Kg}, h, C_L, \mu_a) = -a_{w1}\frac{D}{m} - a_{w2}\frac{L}{m} + g
\end{aligned}
\tag{22}
$$

Accordingly, the altitude constraint is of second order.

For incorporating the altitude constraint in the optimization, the Hamiltonian is changed to yield

$$
H^* = H + \mu(t) G^{(2)}(u_{Kg}, v_{Kg}, w_{Kg}, h, C_L, \mu_a) \tag{23}
$$

with addition of an Lagrange multiplier denoted by $\mu(t)$.

As a consequence, the following relations result

$$
\begin{aligned}
\frac{\mathrm{d}\lambda_u}{\mathrm{d}t} &= -\frac{\partial H}{\partial u_{Kg}} - \mu(t)\frac{\partial}{\partial u_{Kg}} G^{(2)} \\
\frac{\mathrm{d}\lambda_v}{\mathrm{d}t} &= -\frac{\partial H}{\partial v_{Kg}} - \mu(t)\frac{\partial}{\partial v_{Kg}} G^{(2)} \\
\frac{\mathrm{d}\lambda_w}{\mathrm{d}t} &= -\frac{\partial H}{\partial w_{Kg}} - \mu(t)\frac{\partial}{\partial w_{Kg}} G^{(2)} \\
\frac{\mathrm{d}\lambda_h}{\mathrm{d}t} &= -\frac{\partial H}{\partial h} - \mu(t)\frac{\partial}{\partial h} G^{(2)}
\end{aligned}
\tag{24}
$$

The expressions $\mathrm{d}\lambda_x / \mathrm{d}t = 0$ und $\mathrm{d}\lambda_y / \mathrm{d}t = 0$ remain unchanged, because $G^{(2)}$ is independent of x_g and y_g.

For $\mu(t)$, the following relations hold:

a) $\mu(t) = 0$ on the unconstrained arc.

b) $\mu(t) \geq 0$ on the constrained arc. From $\partial H / \partial C_L = 0$ and $\gamma_a = 0$ it follows that

$$
\begin{aligned}
\mu(t) = &-\frac{2kC_L}{\cos \mu_a}(\lambda_u \cos \chi_a + \lambda_v \sin \chi_a) \\
&- \lambda_u \tan \mu_a \sin \chi_a + \lambda_v \tan \mu_a \cos \chi_a - \lambda_w
\end{aligned}
\tag{25}
$$

c) μ can be discontinuous at the entry point of the constrained arc. At the exit point of the constrained arc, μ is continuous. Using a), it is given by $\mu(t_2) = 0$ where t_2 denotes the exit point.

From Eq. (22), the following relation is obtained for the optimal control of the lift coefficient on the constrained arc ($G^{(2)} = 0$ with $\gamma_a = 0$)

$$(C_L)_{opt} = \frac{1}{\cos\mu_a} \frac{mg}{(\rho/2)V^2 S} \tag{26}$$

This relation describes the aerodynamic lift required for an (unsteady) horizontal turn, i.e. $L = mg/\cos\mu_a$.

The optimal bank angle on the constrained arc can be determined, using $\partial H/\partial\mu_a = 0$ with Eqs. (25) and (26), to yield

$$\tan(\mu_a)_{opt} = -\frac{(\rho/2)\ V^2 S}{2kmg} \frac{\lambda_u \sin\chi_a - \lambda_v \cos\chi_a}{\lambda_u \cos\chi_a + \lambda_v \sin\chi_a} \tag{27}$$

In regard to the entry point of the constrained arc, denoted by t_1, the following relations apply:

a) Eqs. (21) and (22) can be used to yield

$$\begin{aligned} h(t_1) &= h_{\min} \\ w_{Kg}(t_1) &= 0 \end{aligned} \tag{28}$$

b) Some of the multipliers are discontinuous. Denoting by t_1^- the time just before the entry point and by t_1^+ immediately after, the following relations hold

$$\begin{aligned} \lambda_h(t_1^+) &= \lambda_h(t_1^-) + \nu_0 \\ \lambda_w(t_1^+) &= \lambda_w(t_1^-) - \nu_1 \end{aligned} \tag{29}$$

where $\nu_0 \geq 0$ und $\nu_1 \geq 0$ are additional unknowns. The other multipliers are continuous at the entry point.

c) The controls are continuous at the entry point. This results from $\nu_1 = \mu(t_1)$ and $H = \text{const}$.

The introduction of switching functions is appropriate for the numerical treatment of constrained arcs. These functions may be specified for the entry and exit points as

$$\begin{aligned} S_1 &:= h(t_1) - h_{\min} = 0 \\ S_2 &:= \mu(t_2) = 0 \end{aligned} \tag{30}$$

In case that the control and state constraints, Eq. (10) and (21), become simultaneously active, the optimal bank angle can be determined with the use of Eq. (26) to yield

$$\cos(\mu_a)_{opt} = \frac{1}{C_{L\max}} \frac{mg}{(\rho/2)^2 S} \tag{31}$$

For the Lagrange multiplier μ, Eqs. (25) remains valid with $C_L = C_{L\max}$ and $\mu_a = (\mu_a)_{opt}$ from Eq. (31).

7. NUMERICAL RESULTS

Numerical results have been achieved treating the optimization of dynamic soaring of seabirds as a boundary value problem. The numerical difficulties in determining optimal dynamic soaring trajectories require powerful computational procedures and efficient computer programs. These problems include the precise treatment of switching conditions, internal point and jump conditions, etc. The computer code applied is based on the method of multiple shooting and provides results of high accuracy (Refs. 27 and 28).

In the numerical treatment, data of an albatross is used as a representative for the seabirds performing dynamic soaring. Reference is made to albatross data given in Refs. 1,12,13. The model data applied in the present investigation are given in Table 1.

	Model Data
m [kg]	9.0
S [m^2]	0.65
b [m]	3.47
C_{D0}	0.033
k	0.019
$(L/D)_{\max}$	20

Table 1 Albatross data

The optimal dynamic soaring cycle which requires minimum wind strength is presented in Fig. 6 which provides a perspective view on its form and shows its extensions in the three dimensions. The dotted arrows denote the begin and end of the optimal cycle, referring to flight conditions of the same energy state and the same direction corresponding to an energy-neutral

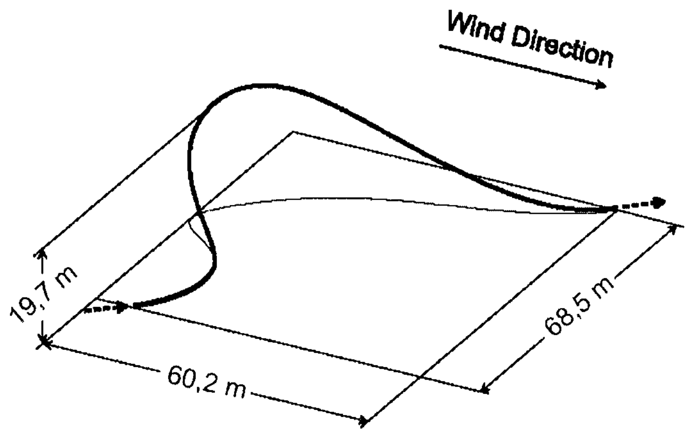

Figure 6. Optimal dynamic soaring cycle requiring minimum wind strength
Optimal cycle time: $(t_{cyc})_{opt} = 7.2$ sec
Wind speed: 9.4 m/s at highest point (h =19.7 m)
5.6 m/s at lowest point ($h_{min} = 0.5$ m)

cycle. As a basic issue, the results concerning the required wind speed and the altitude region compare well with observations and empirical data.

Figs. 7 and 8 present the time histories of state variables, providing more quantitative information about the motion. The altitude range of an optimal cycle, depicted in Fig. 7, extends to about 20 m. The lower altitude limit becomes active for quite a part of the optimal cycle. It is related to the lower curve close to the water surface. The speed behavior is shown in Fig. 8. During the windward climb, the airspeed is larger than the speed relative to the earth while the opposite holds for the leeward descent. The highest speed level is attained in the lower curve.

Results for the optimal controls are presented in Figs. 9 and 10. Fig. 9 shows that the lifting capability is utilized to a large extent. In both the upper and lower curve, the lift coefficient reaches its maximum limit. Correspondingly, the constraint in the lift coefficient becomes active (straight line segments). The unsteady character of dynamic soaring also manifests in the behavior of the bank angle the time history of which is shown in Fig. 10. The greatest bank angle amounts to about 75 deg.

8. CONCLUSIONS

Dynamic soaring which is a flight technique of seabirds for extracting energy from horizontally moving air in a shear flow is treated as an optimal

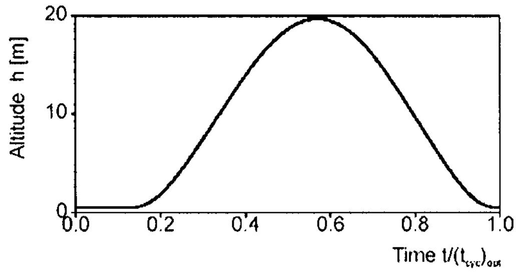

Figure 7. Altitude

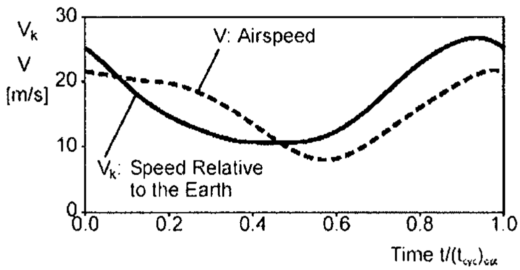

Figure 8. Speeds

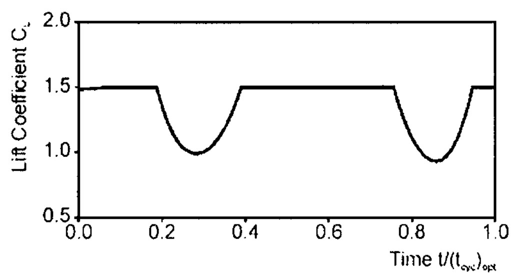

Figure 9. Lift coefficient

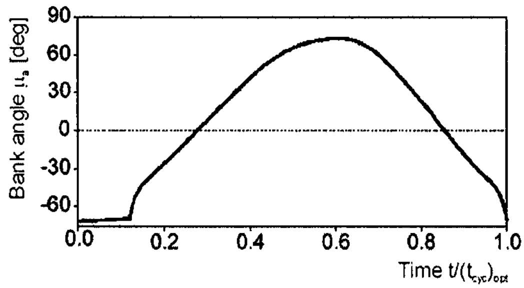

Figure 10. Bank Angle

control problem. It involves a rather complex and highly dynamic maneuver consisting of a sequence of climbing, turning and descending flight phases in order to achieve an energy gain from the shear wind which exists in an altitude layer above the water surface. The basic objective of this paper is to determine the minimum wind strength required for dynamic soaring of seabirds, using optimal control theory. Mathematical models for the motion of a bird in horizontally moving air and for the shear wind are presented. Optimization considerations are applied based on the minimum principle, yielding necessary optimality conditions. Furthermore, switching conditions are introduced in order to deal with an altitude constraint of the dynamic soaring trajectory. Numerical results of high accuracy are generated using an efficient computational procedure based on the method of multiple shooting. The results which concern an albatross as a representative for seabirds performing dynamic soaring concern the minimum shear wind gradient and properties of the related optimal trajectory.

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